

Should we discount tomorrow?

Daily decision support for perishable food

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A category manager must choose tomorrow's discount. We use public sales histories to compare supported price options while limiting loss of predicted sales value.

01 Why these five stores?

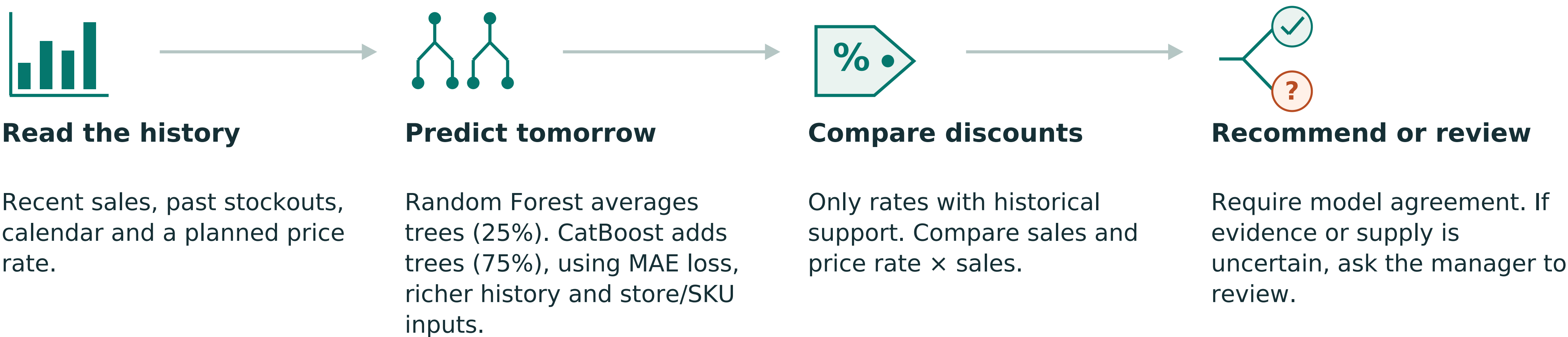
Use products with enough discount variation to study, then choose the stores with the most eligible products.



Eligibility: 15-85% discounted days, 6+ discount-state changes, and fewer than 20% zero-sales days.

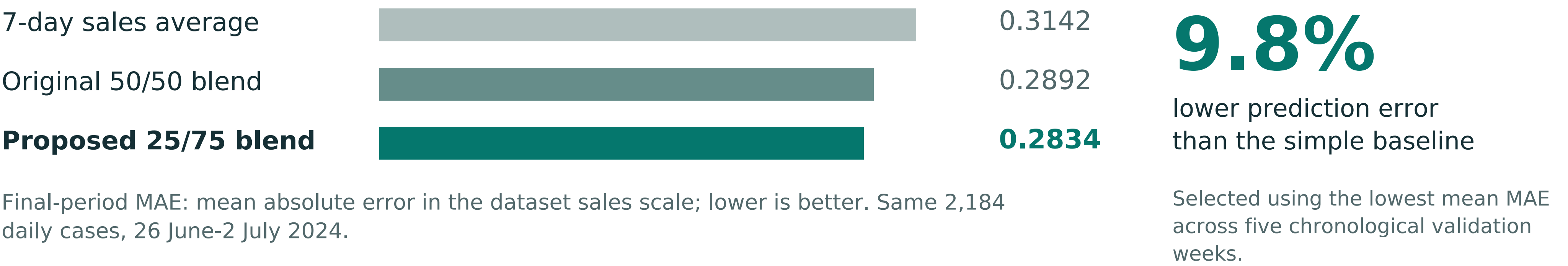
Selected stores: 343 / 18 / 235 / 182 / 154. One city; chosen retrospectively using the full training period.

02 Forecast first. Then make a cautious decision.



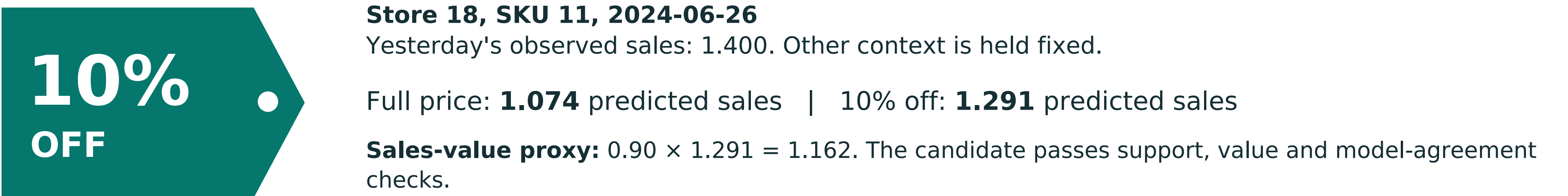
The rule: seek at least 5% more predicted sales while keeping at least 95% of full-price sales value. These are decision thresholds, not guarantees.

03 More accurate on the five-store benchmark



One technical experiment: keep the extended features fixed and change CatBoost's training loss. MAE loss reduced validation MAE from **0.2972 to 0.2945** versus RMSE loss.

04 What would the manager actually see?



An illustrative model scenario, not an observed response to changing the price.

Where evidence stops: on 700 cases from 93 new stores, MAE was 0.2890 vs. 0.2884 for the original blend: no improvement; the models were not retuned. The main benchmark was already viewed; 96.9% of cases without recent stockouts still recommend discounts. Historical discounts are not causal evidence, and stockouts do not reveal inventory age or waste.